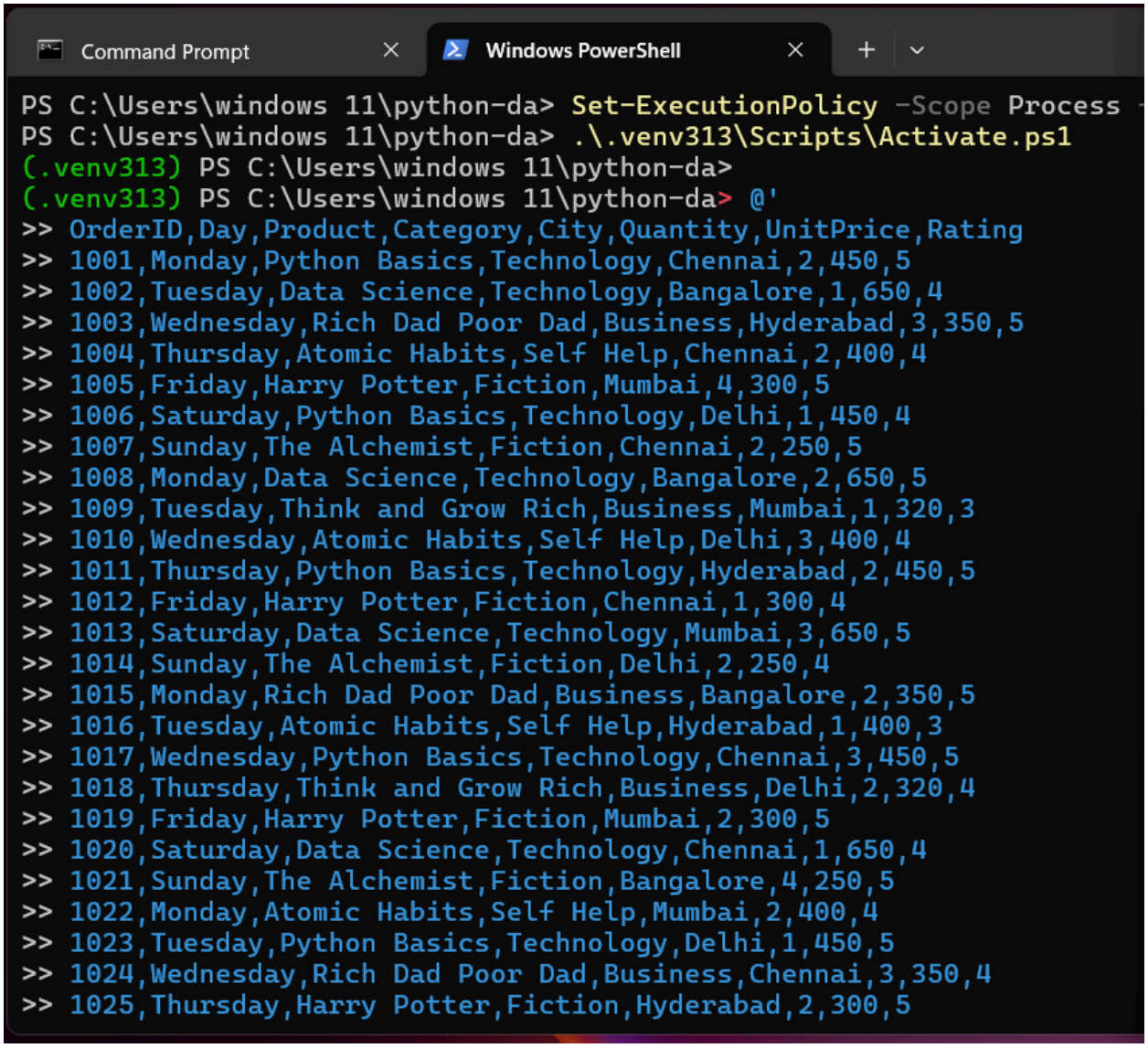
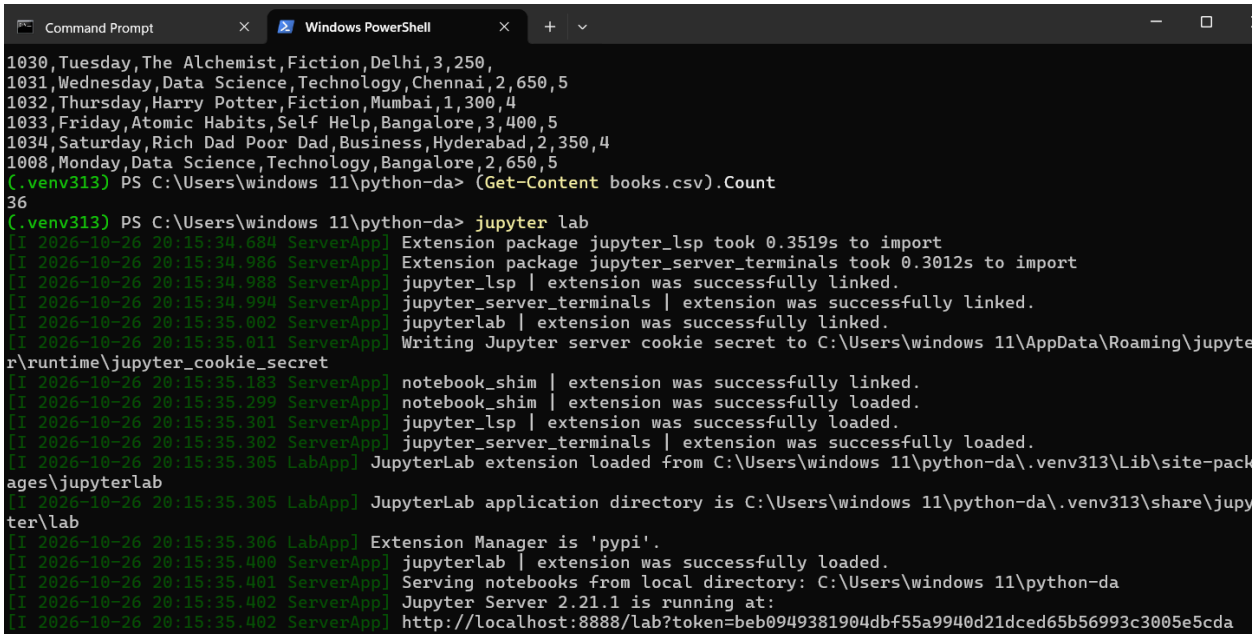
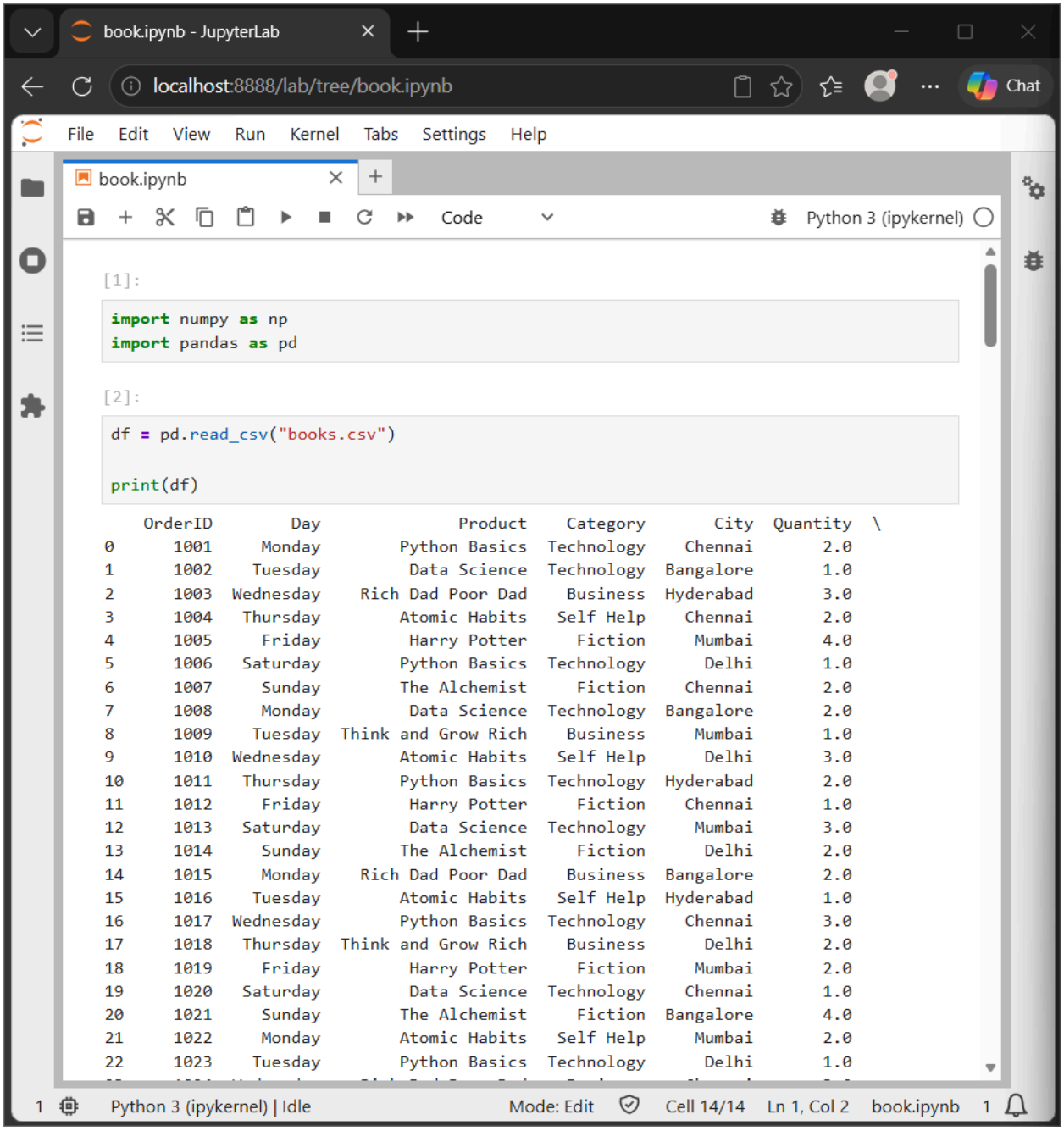
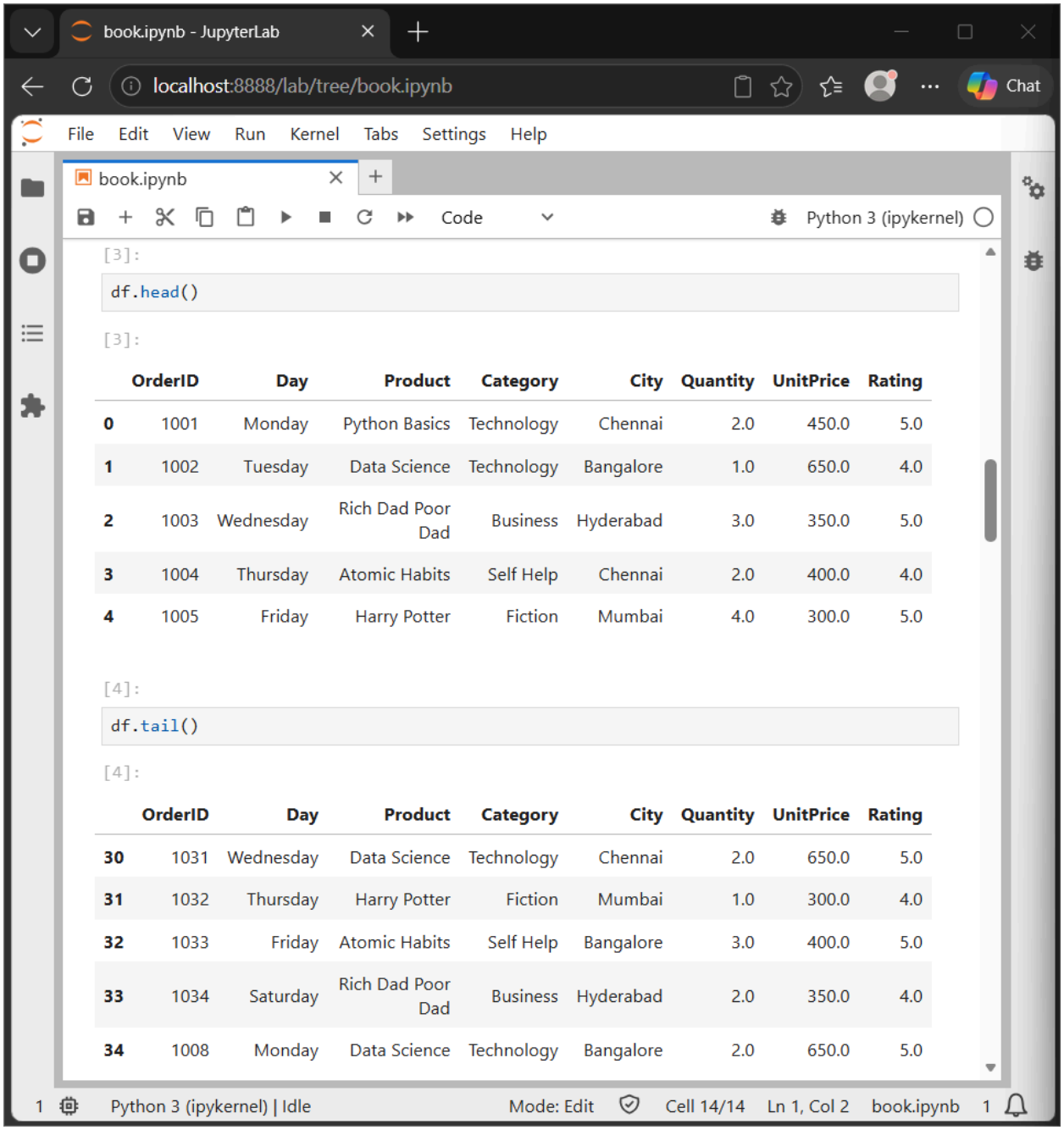

Command Prompt
Windows PowerShell
PS C:\Users\windows 11\python-da> Set-ExecutionPolicy -Scope Process
PS C:\Users\windows 11\python-da> .\.venv313\Scripts\Activate.ps1
(.venv313) PS C:\Users\windows 11\python-da>
(.venv313) PS C:\Users\windows 11\python-da> @'
>> OrderID,Day,Product,Category,City,Quantity,UnitPrice,Rating
>> 1001,Monday,Python Basics,Technology,Chennai,2,450,5
>> 1002,Tuesday,Data Science,Technology,Bangalore,1,650,4
>> 1003,Wednesday,Rich Dad Poor Dad,Business,Hyderabad,3,350,5
>> 1004,Thursday,Atomic Habits,Self Help,Chennai,2,400,4
>> 1005,Friday,Harry Potter,Fiction,Mumbai,4,300,5
>> 1006,Saturday,Python Basics,Technology,Delhi,1,450,4
>> 1007,Sunday,The Alchemist,Fiction,Chennai,2,250,5
>> 1008,Monday,Data Science,Technology,Bangalore,2,650,5
>> 1009,Tuesday,Think and Grow Rich,Business,Mumbai,1,320,3
>> 1010,Wednesday,Atomic Habits,Self Help,Delhi,3,400,4
>> 1011,Thursday,Python Basics,Technology,Hyderabad,2,450,5
>> 1012,Friday,Harry Potter,Fiction,Chennai,1,300,4
>> 1013,Saturday,Data Science,Technology,Mumbai,3,650,5
>> 1014,Sunday,The Alchemist,Fiction,Delhi,2,250,4
>> 1015,Monday,Rich Dad Poor Dad,Business,Bangalore,2,350,5
>> 1016,Tuesday,Atomic Habits,Self Help,Hyderabad,1,400,3
>> 1017,Wednesday,Python Basics,Technology,Chennai,3,450,5
>> 1018,Thursday,Think and Grow Rich,Business,Delhi,2,320,4
>> 1019,Friday,Harry Potter,Fiction,Mumbai,2,300,5
>> 1020,Saturday,Data Science,Technology,Chennai,1,650,4
>> 1021,Sunday,The Alchemist,Fiction,Bangalore,4,250,5
>> 1022,Monday,Atomic Habits,Self Help,Mumbai,2,400,4
>> 1023,Tuesday,Python Basics,Technology,Delhi,1,450,5
>> 1024,Wednesday,Rich Dad Poor Dad,Business,Chennai,3,350,4
>> 1025,Thursday,Harry Potter,Fiction,Hyderabad,2,300,5

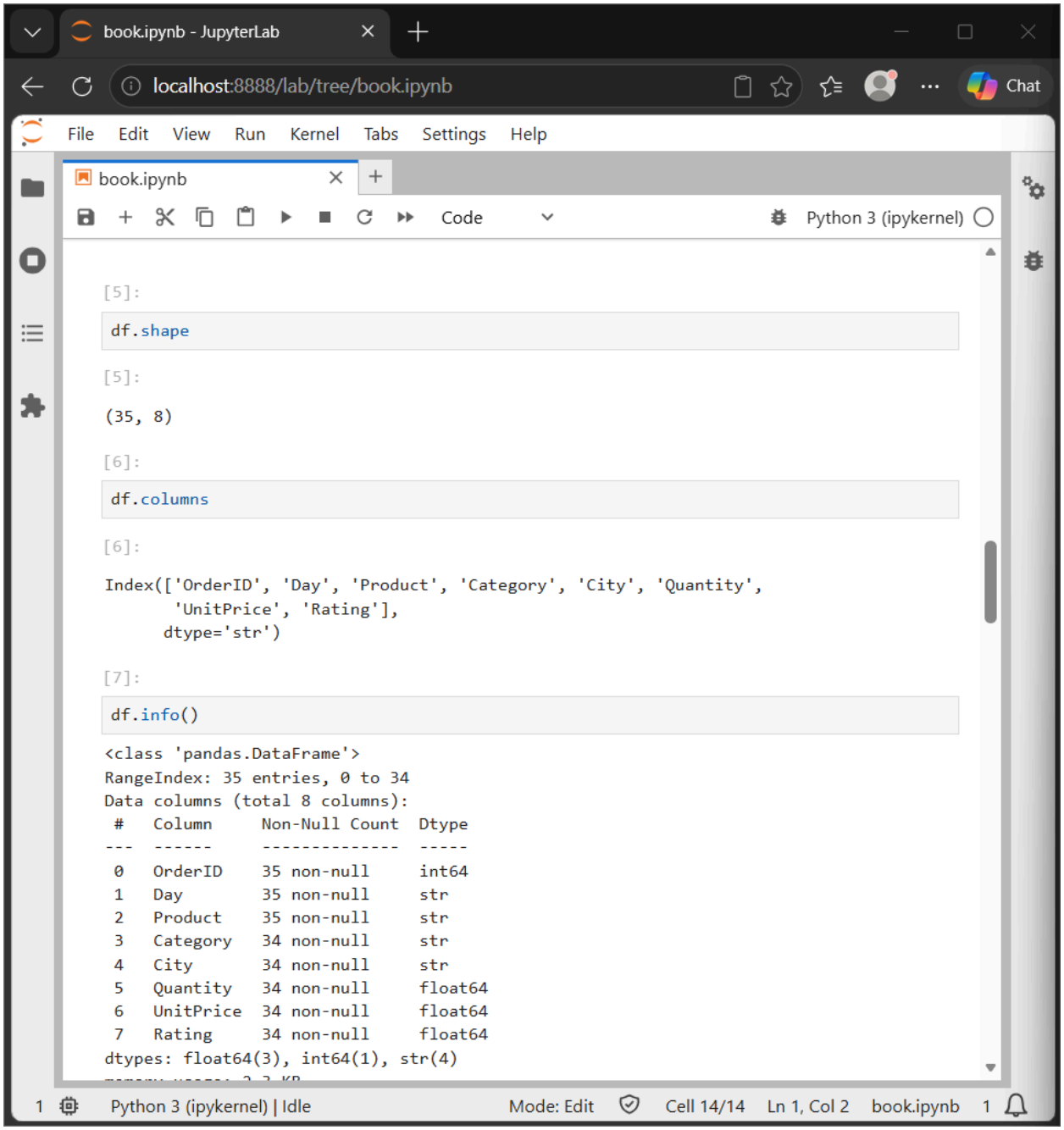

Command Prompt
Windows PowerShell
1030,Tuesday,The Alchemist,Fiction,Delhi,3,250,
1031,Wednesday,Data Science,Technology,Chennai,2,650,5
1032,Thursday,Harry Potter,Fiction,Mumbai,1,300,4
1033,Friday,Atomic Habits,Self Help,Bangalore,3,400,5
1034,Saturday,Rich Dad Poor Dad,Business,Hyderabad,2,350,4
1008,Monday,Data Science,Technology,Bangalore,2,650,5
(.venv313) PS C:\Users\windows 11\python-da> (Get-Content books.csv).Count
36
(.venv313) PS C:\Users\windows 11\python-da> jupyter lab
[I 2026-10-26 20:15:34.684 ServerApp] Extension package jupyter_lsp took 0.3519s to import
[I 2026-10-26 20:15:34.986 ServerApp] Extension package jupyter_server_terminals took 0.3012s to import
[I 2026-10-26 20:15:34.988 ServerApp] jupyter_lsp | extension was successfully linked.
[I 2026-10-26 20:15:34.994 ServerApp] jupyter_server_terminals | extension was successfully linked.
[I 2026-10-26 20:15:35.002 ServerApp] jupyterlab | extension was successfully linked.
[I 2026-10-26 20:15:35.011 ServerApp] Writing Jupyter server cookie secret to C:\Users\windows 11\AppData\Roaming\jupyte
r\runtime\jupyter_cookie_secret
[I 2026-10-26 20:15:35.183 ServerApp] notebook_shim | extension was successfully linked.
[I 2026-10-26 20:15:35.299 ServerApp] notebook_shim | extension was successfully loaded.
[I 2026-10-26 20:15:35.301 ServerApp] jupyter_lsp | extension was successfully loaded.
[I 2026-10-26 20:15:35.302 ServerApp] jupyter_server_terminals | extension was successfully loaded.
[I 2026-10-26 20:15:35.305 LabApp] JupyterLab extension loaded from C:\Users\windows 11\python-da\.venv313\Lib\site-pack
ages\jupyterlab
[I 2026-10-26 20:15:35.305 LabApp] JupyterLab application directory is C:\Users\windows 11\python-da\.venv313\share\jupy
ter\lab
[I 2026-10-26 20:15:35.306 LabApp] Extension Manager is 'pypi'.
[I 2026-10-26 20:15:35.400 ServerApp] jupyterlab | extension was successfully loaded.
[I 2026-10-26 20:15:35.401 ServerApp] Serving notebooks from local directory: C:\Users\windows 11\python-da
[I 2026-10-26 20:15:35.402 ServerApp] Jupyter Server 2.21.1 is running at:
[I 2026-10-26 20:15:35.402 ServerApp] http://localhost:8888/lab?token=beb0949381904dbf55a9940d21dced65b56993c3005e5cda

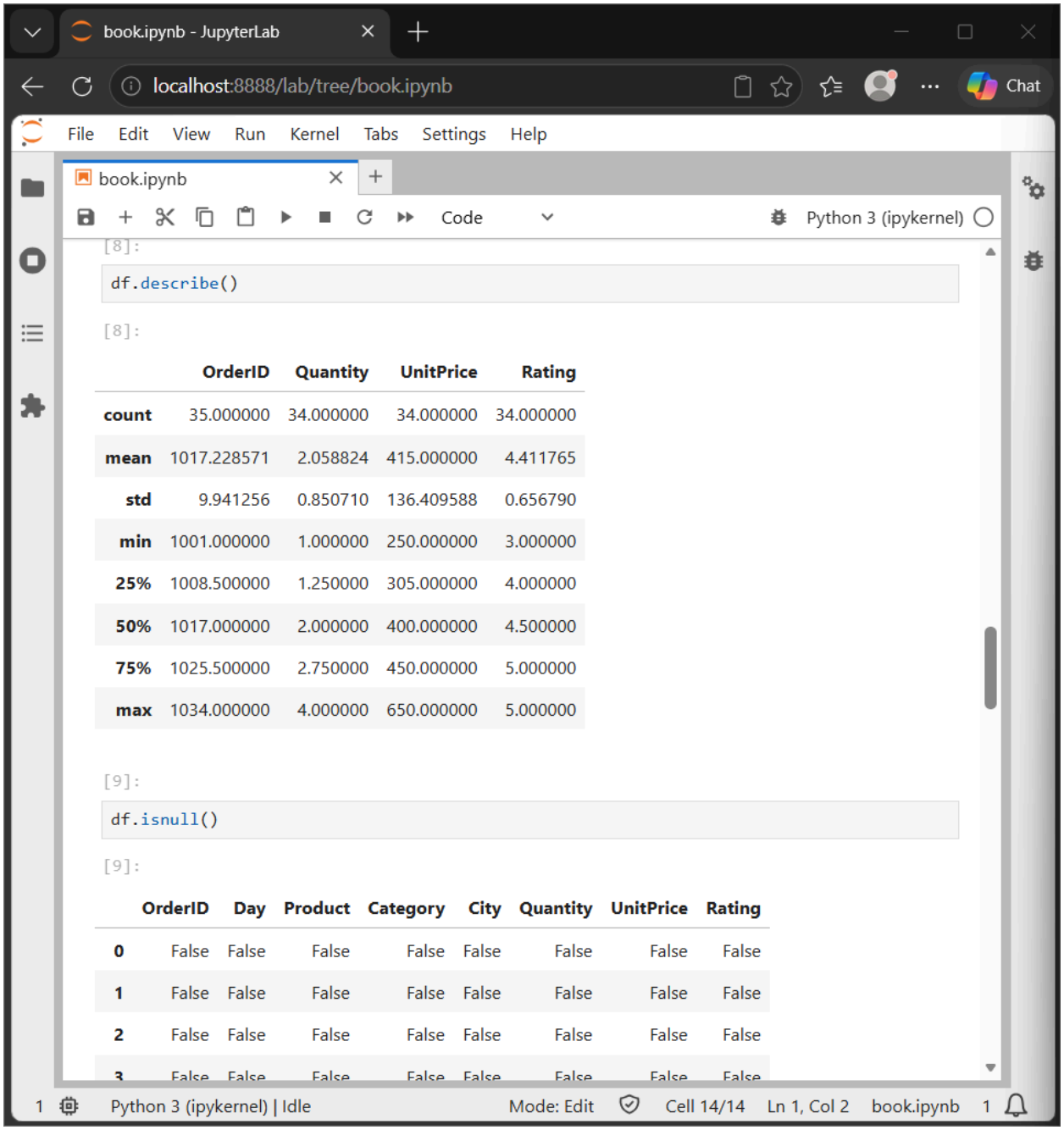

book.ipynb - JupyterLab
localhost:8888/lab/tree/book.ipynb
Chat
File Edit View Run Kernel Tabs Settings Help
book.ipynb
Code
Python 3 (ipykernel)
[1]:
import numpy as np
import pandas as pd
[2]:
df = pd.read_csv("books.csv")
print(df)
OrderID Day Product Category City Quantity \
0 1001 Monday Python Basics Technology Chennai 2.0
1 1002 Tuesday Data Science Technology Bangalore 1.0
2 1003 Wednesday Rich Dad Poor Dad Business Hyderabad 3.0
3 1004 Thursday Atomic Habits Self Help Chennai 2.0
4 1005 Friday Harry Potter Fiction Mumbai 4.0
5 1006 Saturday Python Basics Technology Delhi 1.0
6 1007 Sunday The Alchemist Fiction Chennai 2.0
7 1008 Monday Data Science Technology Bangalore 2.0
8 1009 Tuesday Think and Grow Rich Business Mumbai 1.0
9 1010 Wednesday Atomic Habits Self Help Delhi 3.0
10 1011 Thursday Python Basics Technology Hyderabad 2.0
11 1012 Friday Harry Potter Fiction Chennai 1.0
12 1013 Saturday Data Science Technology Mumbai 3.0
13 1014 Sunday The Alchemist Fiction Delhi 2.0
14 1015 Monday Rich Dad Poor Dad Business Bangalore 2.0
15 1016 Tuesday Atomic Habits Self Help Hyderabad 1.0
16 1017 Wednesday Python Basics Technology Chennai 3.0
17 1018 Thursday Think and Grow Rich Business Delhi 2.0
18 1019 Friday Harry Potter Fiction Mumbai 2.0
19 1020 Saturday Data Science Technology Chennai 1.0
20 1021 Sunday The Alchemist Fiction Bangalore 4.0
21 1022 Monday Atomic Habits Self Help Mumbai 2.0
22 1023 Tuesday Python Basics Technology Delhi 1.0
1 Python 3 (ipykernel) | Idle Mode: Edit Cell 14/14 Ln 1, Col 2 book.ipynb 1

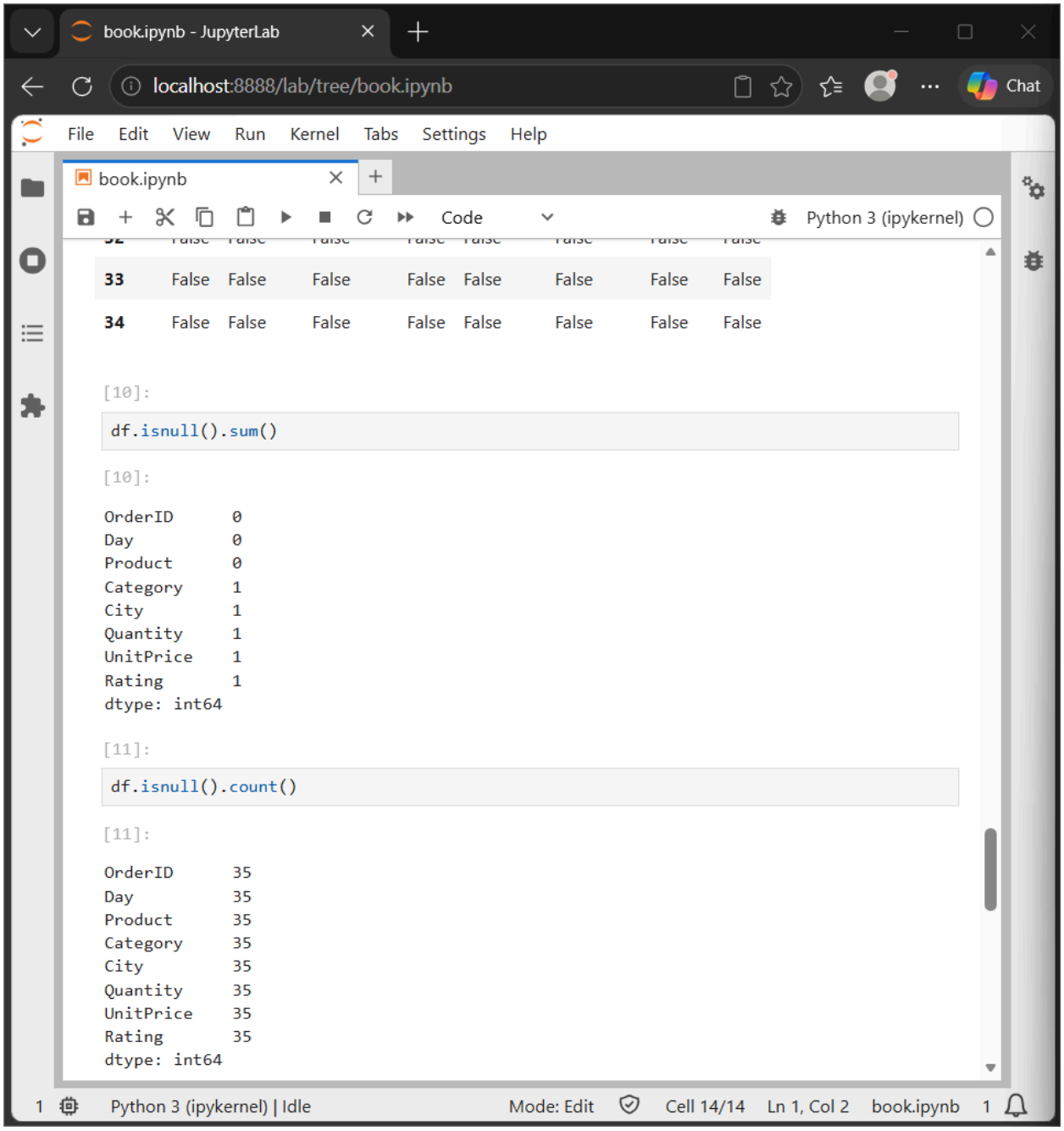


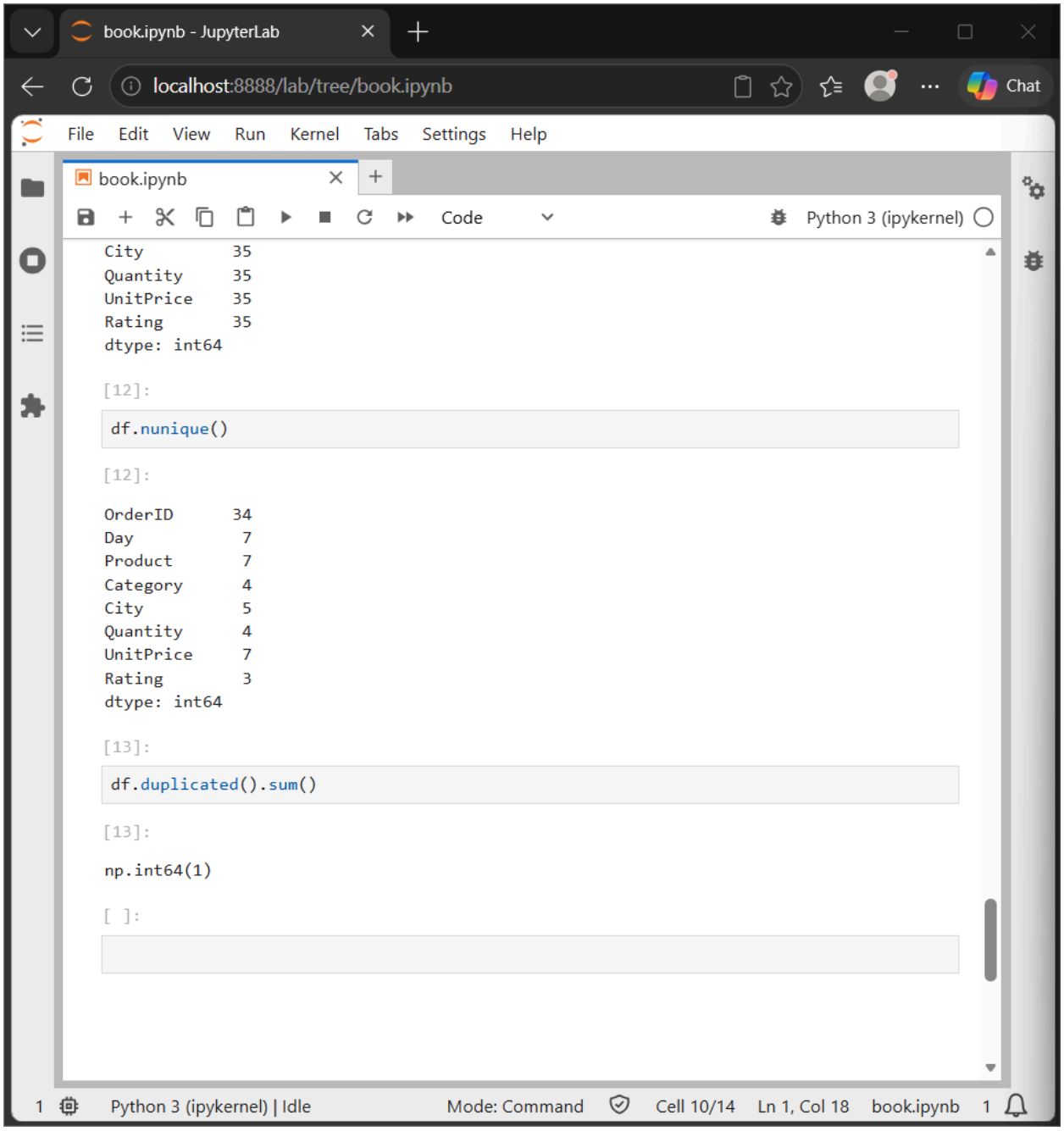
	OrderID	Day	Product	Category	City	Quantity	UnitPrice	Rating
0	1001	Monday	Python Basics	Technology	Chennai	2.0	450.0	5.0
1	1002	Tuesday	Data Science	Technology	Bangalore	1.0	650.0	4.0
2	1003	Wednesday	Rich Dad Poor Dad	Business	Hyderabad	3.0	350.0	5.0
3	1004	Thursday	Atomic Habits	Self Help	Chennai	2.0	400.0	4.0
4	1005	Friday	Harry Potter	Fiction	Mumbai	4.0	300.0	5.0

	OrderID	Day	Product	Category	City	Quantity	UnitPrice	Rating
30	1031	Wednesday	Data Science	Technology	Chennai	2.0	650.0	5.0
31	1032	Thursday	Harry Potter	Fiction	Mumbai	1.0	300.0	4.0
32	1033	Friday	Atomic Habits	Self Help	Bangalore	3.0	400.0	5.0
33	1034	Saturday	Rich Dad Poor Dad	Business	Hyderabad	2.0	350.0	4.0
34	1008	Monday	Data Science	Technology	Bangalore	2.0	650.0	5.0

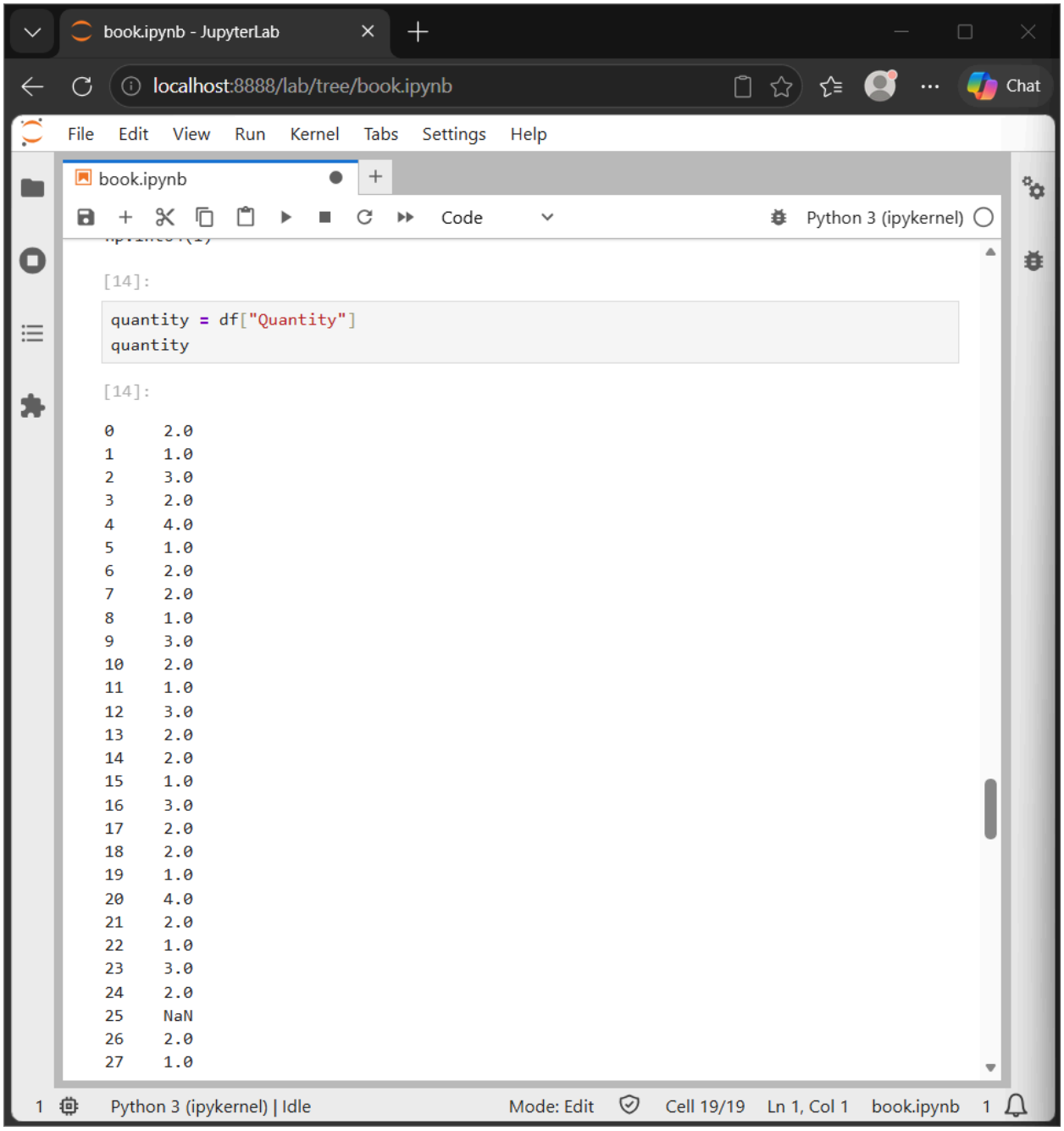

book.ipynb - JupyterLab
localhost:8888/lab/tree/book.ipynb
Chat
File Edit View Run Kernel Tabs Settings Help
book.ipynb
Code
Python 3 (ipykernel)
[5]:
df.shape
[5]:
(35, 8)
[6]:
df.columns
[6]:
Index(['OrderID', 'Day', 'Product', 'Category', 'City', 'Quantity',
'UnitPrice', 'Rating'],
dtype='str')
[7]:
df.info()
<class 'pandas.DataFrame'>
RangeIndex: 35 entries, 0 to 34
Data columns (total 8 columns):
Column Non-Null Count Dtype
--- ------ -------------- -----
0 OrderID 35 non-null int64
1 Day 35 non-null str
2 Product 35 non-null str
3 Category 34 non-null str
4 City 34 non-null str
5 Quantity 34 non-null float64
6 UnitPrice 34 non-null float64
7 Rating 34 non-null float64
dtypes: float64(3), int64(1), str(4)
Python 3 (ipykernel) | Idle
Mode: Edit
Cell 14/14
Ln 1, Col 2
book.ipynb

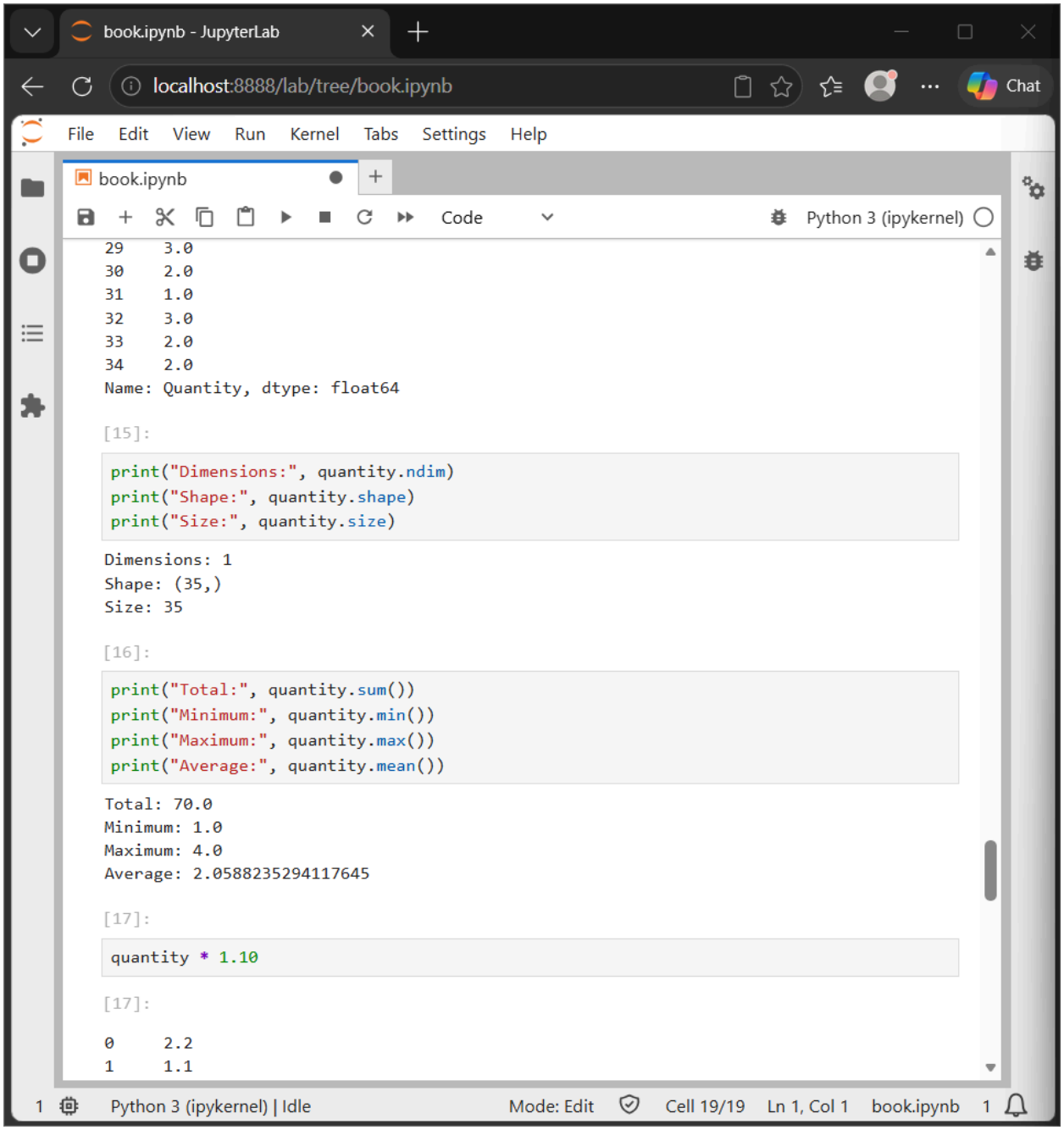

book.ipynb - JupyterLab
localhost:8888/lab/tree/book.ipynb
Chat
File Edit View Run Kernel Tabs Settings Help
book.ipynb
Code
Python 3 (ipykernel)
[8]:
df.describe()
[8]:
OrderID Quantity UnitPrice Rating
count 35.000000 34.000000 34.000000 34.000000
mean 1017.228571 2.058824 415.000000 4.411765
std 9.941256 0.850710 136.409588 0.656790
min 1001.000000 1.000000 250.000000 3.000000
25% 1008.500000 1.250000 305.000000 4.000000
50% 1017.000000 2.000000 400.000000 4.500000
75% 1025.500000 2.750000 450.000000 5.000000
max 1034.000000 4.000000 650.000000 5.000000
[9]:
df.isnull()
[9]:
OrderID Day Product Category City Quantity UnitPrice Rating
0 False False False False False False False False
1 False False False False False False False False
2 False False False False False False False False
3 False False False False False False False False
1 Python 3 (ipykernel) | Idle Mode: Edit Cell 14/14 Ln 1, Col 2 book.ipynb 1

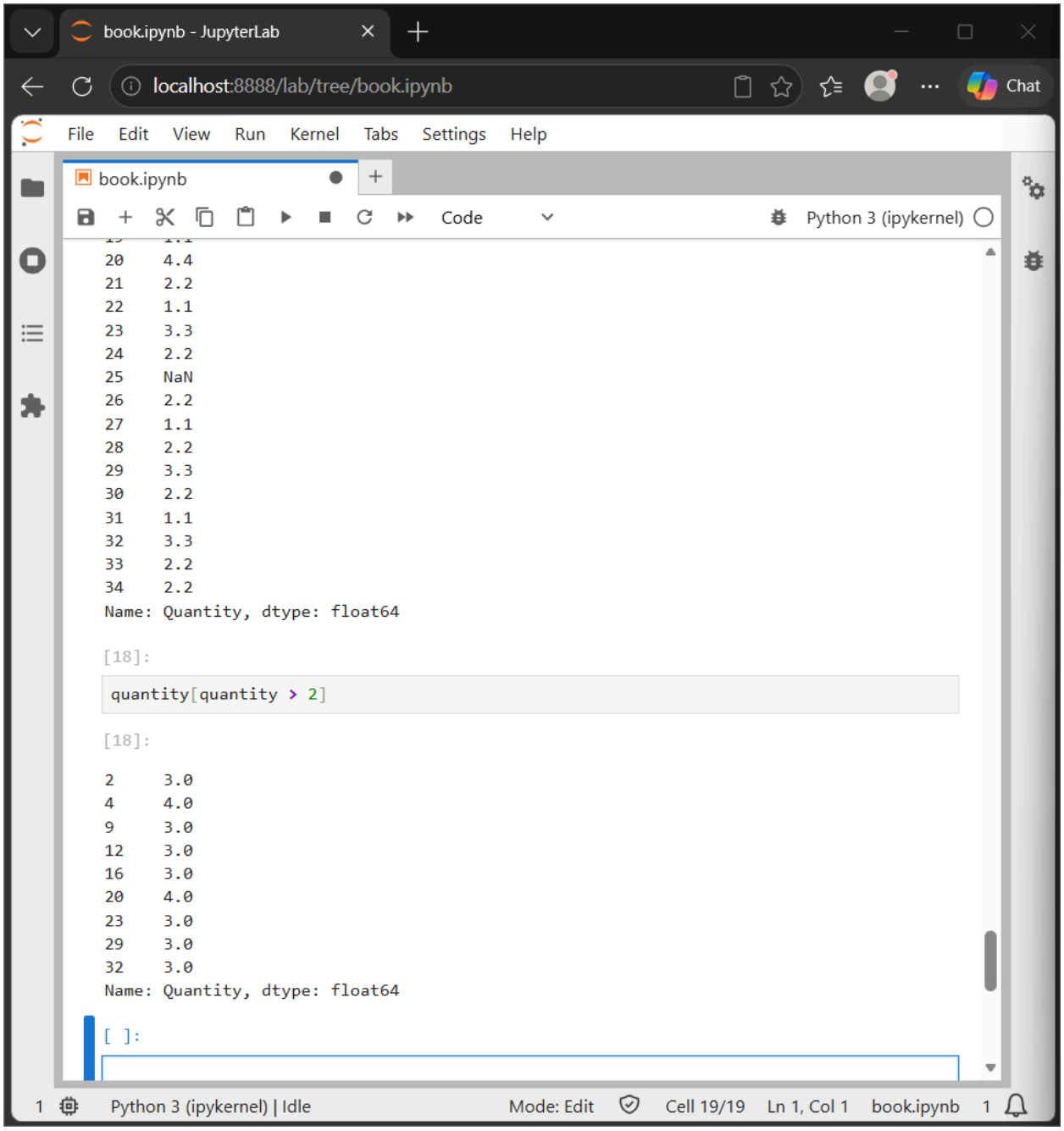

book.ipynb - JupyterLab
localhost:8888/lab/tree/book.ipynb
Chat
File Edit View Run Kernel Tabs Settings Help
book.ipynb
Code
Python 3 (ipykernel)
33 False False False False False False False False
34 False False False False False False False False
[10]:
df.isnull().sum()
[10]:
OrderID 0
Day 0
Product 0
Category 1
City 1
Quantity 1
UnitPrice 1
Rating 1
dtype: int64
[11]:
df.isnull().count()
[11]:
OrderID 35
Day 35
Product 35
Category 35
City 35
Quantity 35
UnitPrice 35
Rating 35
dtype: int64
1 Python 3 (ipykernel) | Idle
Mode: Edit
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book.ipynb
1



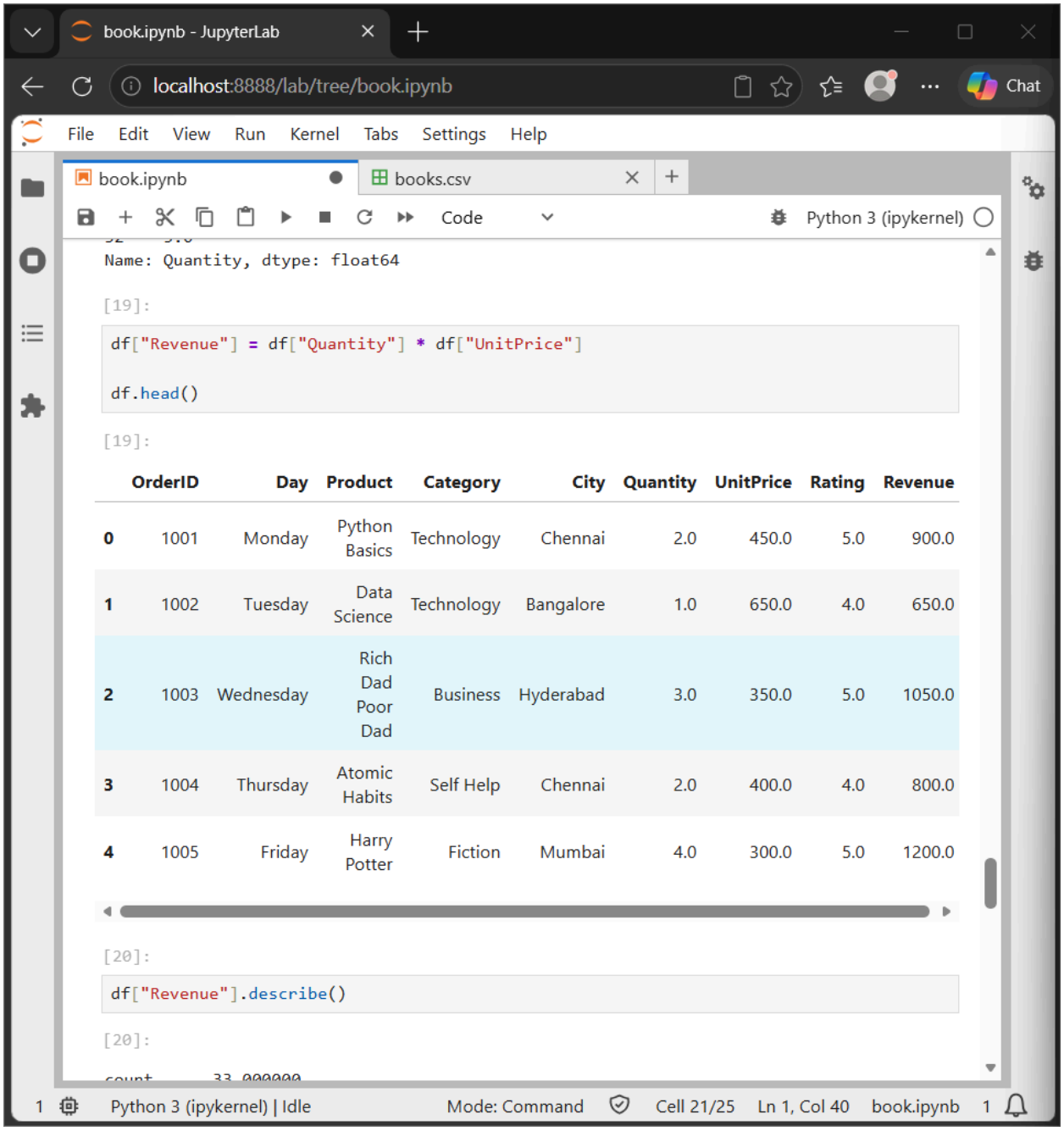
Pandas Series :

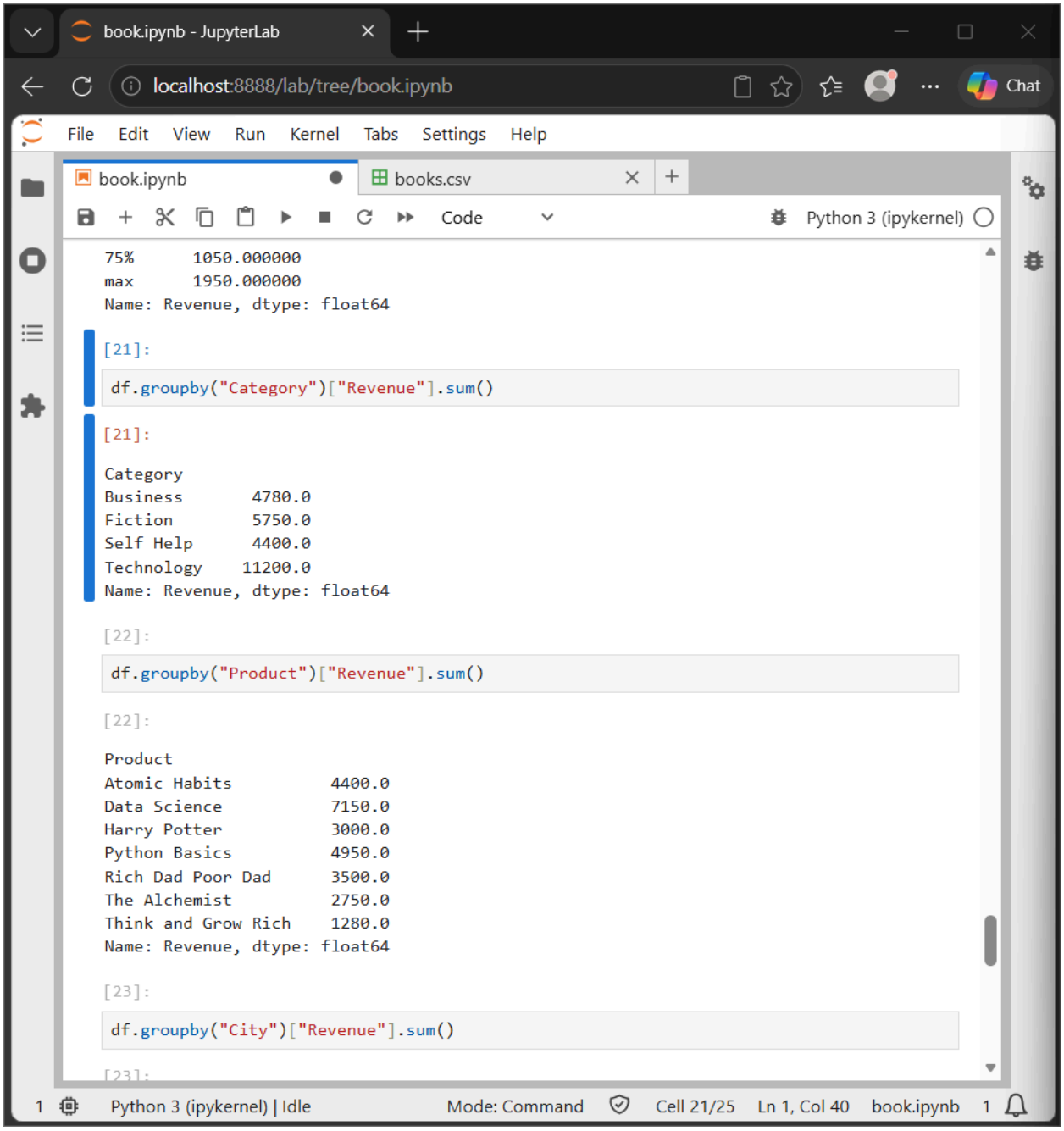

book.ipynb - JupyterLab
localhost:8888/lab/tree/book.ipynb
Chat
File Edit View Run Kernel Tabs Settings Help
book.ipynb
Code
Python 3 (ipykernel)
[14]:
quantity = df["Quantity"]
quantity
[14]:
0 2.0
1 1.0
2 3.0
3 2.0
4 4.0
5 1.0
6 2.0
7 2.0
8 1.0
9 3.0
10 2.0
11 1.0
12 3.0
13 2.0
14 2.0
15 1.0
16 3.0
17 2.0
18 2.0
19 1.0
20 4.0
21 2.0
22 1.0
23 3.0
24 2.0
25 NaN
26 2.0
27 1.0
1 Python 3 (ipykernel) | Idle
Mode: Edit
Cell 19/19
Ln 1, Col 1
book.ipynb
1

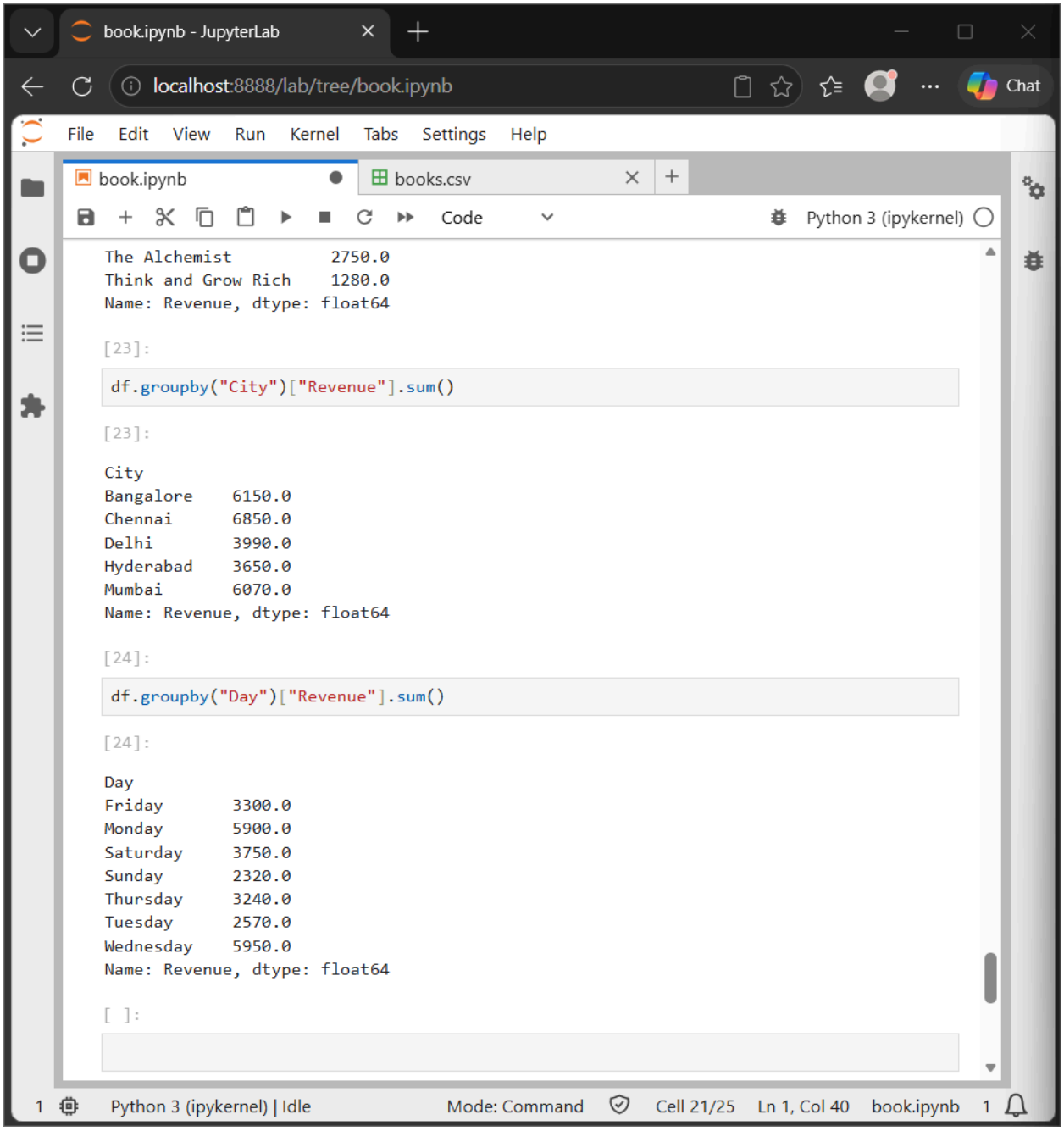

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book.ipynb
Code
Python 3 (ipykernel)
29 3.0
30 2.0
31 1.0
32 3.0
33 2.0
34 2.0
Name: Quantity, dtype: float64
[15]:
print("Dimensions:", quantity.ndim)
print("Shape:", quantity.shape)
print("Size:", quantity.size)
Dimensions: 1
Shape: (35,)
Size: 35
[16]:
print("Total:", quantity.sum())
print("Minimum:", quantity.min())
print("Maximum:", quantity.max())
print("Average:", quantity.mean())
Total: 70.0
Minimum: 1.0
Maximum: 4.0
Average: 2.0588235294117645
[17]:
quantity * 1.10
[17]:
0 2.2
1 1.1
1 Python 3 (ipykernel) | Idle Mode: Edit Cell 19/19 Ln 1, Col 1 book.ipynb 1

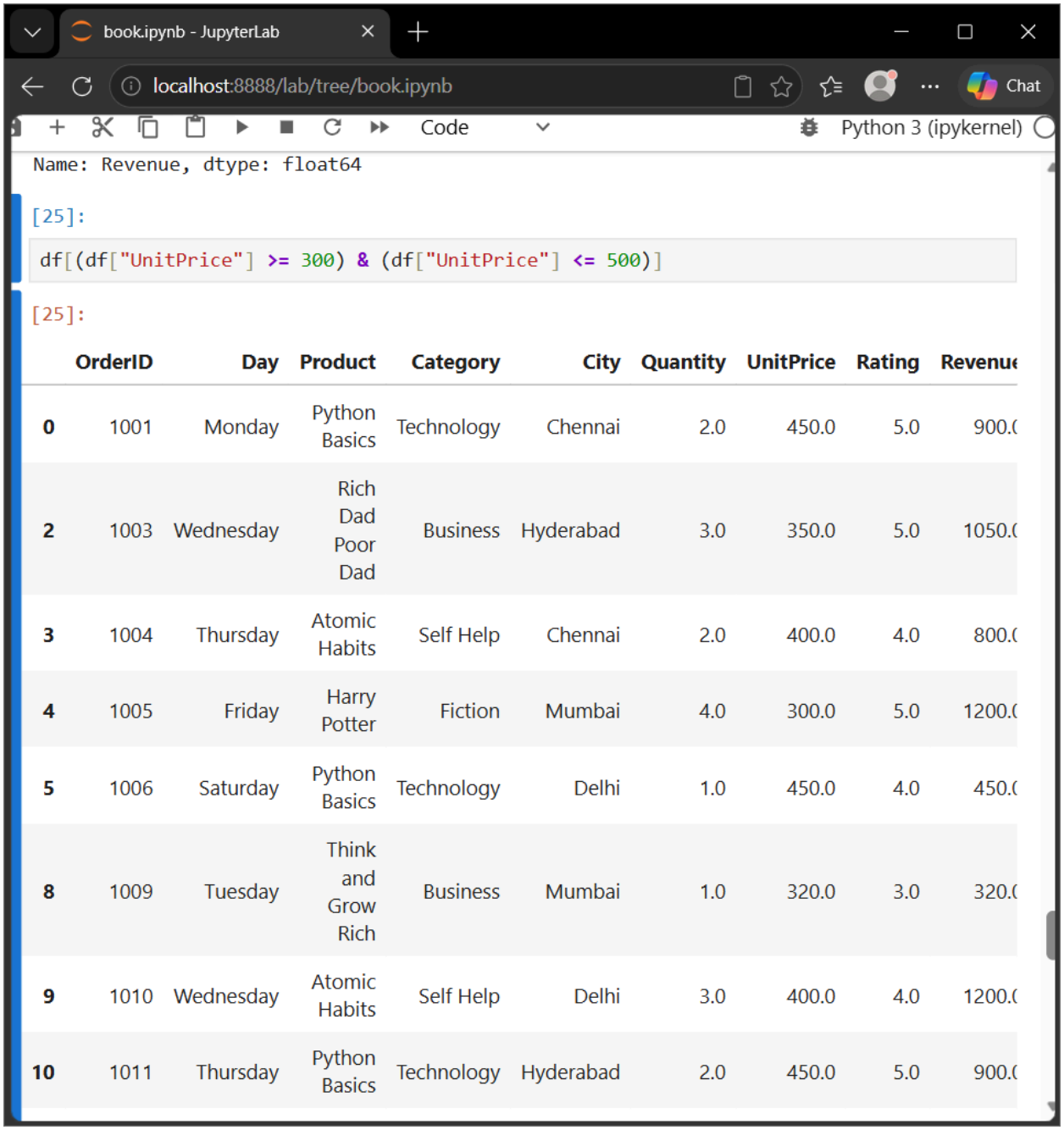


Creating a “Revenue” Column


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book.ipynb
books.csv
Code
Python 3 (ipykernel)
Name: Quantity, dtype: float64
[19]:
df["Revenue"] = df["Quantity"] * df["UnitPrice"]
df.head()
[19]:
OrderID Day Product Category City Quantity UnitPrice Rating Revenue
0 1001 Monday Python Basics Technology Chennai 2.0 450.0 5.0 900.0
1 1002 Tuesday Data Science Technology Bangalore 1.0 650.0 4.0 650.0
2 1003 Wednesday Rich Dad Poor Dad Business Hyderabad 3.0 350.0 5.0 1050.0
3 1004 Thursday Atomic Habits Self Help Chennai 2.0 400.0 4.0 800.0
4 1005 Friday Harry Potter Fiction Mumbai 4.0 300.0 5.0 1200.0
[20]:
df["Revenue"].describe()
[20]:
Python 3 (ipykernel) | Idle
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book.ipynb
books.csv
Code
Python 3 (ipykernel)
75% 1050.000000
max 1950.000000
Name: Revenue, dtype: float64
[21]:
df.groupby("Category")["Revenue"].sum()
[21]:
Category
Business 4780.0
Fiction 5750.0
Self Help 4400.0
Technology 11200.0
Name: Revenue, dtype: float64
[22]:
df.groupby("Product")["Revenue"].sum()
[22]:
Product
Atomic Habits 4400.0
Data Science 7150.0
Harry Potter 3000.0
Python Basics 4950.0
Rich Dad Poor Dad 3500.0
The Alchemist 2750.0
Think and Grow Rich 1280.0
Name: Revenue, dtype: float64
[23]:
df.groupby("City")["Revenue"].sum()
[23]:
1 Python 3 (ipykernel) | Idle Mode: Command Cell 21/25 Ln 1, Col 40 book.ipynb 1


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book.ipynb
books.csv
Code
Python 3 (ipykernel)
The Alchemist 2750.0
Think and Grow Rich 1280.0
Name: Revenue, dtype: float64
[23]:
df.groupby("City")["Revenue"].sum()
[23]:
City
Bangalore 6150.0
Chennai 6850.0
Delhi 3990.0
Hyderabad 3650.0
Mumbai 6070.0
Name: Revenue, dtype: float64
[24]:
df.groupby("Day")["Revenue"].sum()
[24]:
Day
Friday 3300.0
Monday 5900.0
Saturday 3750.0
Sunday 2320.0
Thursday 3240.0
Tuesday 2570.0
Wednesday 5950.0
Name: Revenue, dtype: float64
[ ]:
1 Python 3 (ipykernel) | Idle
Mode: Command
Cell 21/25
Ln 1, Col 40
book.ipynb
1



	OrderID	Day	Product	Category	City	Quantity	UnitPrice	Rating	Revenue
0	1001	Monday	Python Basics	Technology	Chennai	2.0	450.0	5.0	900.(
2	1003	Wednesday	Rich Dad Poor Dad	Business	Hyderabad	3.0	350.0	5.0	1050.(
3	1004	Thursday	Atomic Habits	Self Help	Chennai	2.0	400.0	4.0	800.(
4	1005	Friday	Harry Potter	Fiction	Mumbai	4.0	300.0	5.0	1200.(
5	1006	Saturday	Python Basics	Technology	Delhi	1.0	450.0	4.0	450.(
8	1009	Tuesday	Think and Grow Rich	Business	Mumbai	1.0	320.0	3.0	320.(
9	1010	Wednesday	Atomic Habits	Self Help	Delhi	3.0	400.0	4.0	1200.(
10	1011	Thursday	Python Basics	Technology	Hyderabad	2.0	450.0	5.0	900.(